Bernardo Bahia Monteiro

(734) 205 8479 – bbahia@umich.edu –

EDUCATION

University of Michigan — Ann Arbor, MI
PhD Aerospace Engineering, MSE Aerospace Engineering — GPA: 4.0 — September 2019–May 2024 (expected)
Advisors: Carlos E. S. Cesnik, Ilya Kolmanovsky
Research interest: Develop control-related metrics for Multidisciplinary Design Optimization (MDO) that can estimate the closed-loop performance of the aircraft without control design
Projects: Airbus-Michigan Center for Aero-Servo-Elasticity of Very Flexible Aircraft (CASE-VFA, 2019–2022)
NASA's Supersonic Configurations at Low Speeds (SCALOS, 2022–2023)

Universidade Federal de Minas Gerais — Belo Horizonte, Brazil
BSc Aerospace Engineering — GPA: 4.9/5.0 — February 2014–June 2019
Admitted in first place; graduated first in class
Research assistant under Prof. Ricardo Utsch (2015–2018, numerical analysis)
Model rocketry team

University of Strathclyde — Glasgow, UK
One semester exchange in Aero-mechanical Engineering — January–July, 2017
First class result in all disciplines, mostly at the master's level
Summer research assistant for the "Future UK Small Payload Launcher" project:
- Feasibility study of a horizontal rocket launcher with a reusable first stage
- Multidisciplinary Design Optimization and Optimal Control

WORK EXPERIENCE

Embraer — São José dos Campos, Brazil
Flight Operations Engineering — January, 2018–January, 2019
- Design and implement software solutions to meet customers' demands (Java/C)
  - e.g., calculate friction coefficient of a contaminated runway from flight data
- Support meetings with airline customers
- Formulate and develop proof-of-concept solutions for flight data analysis (Python)

Preliminary Design — February, 2019–June, 2019
- Program semi-empirical aerodynamic models adapted for the in-house MDO code (Fortran)
- Attend design reviews

PUBLICATIONS

[1] Bahia Monteiro, B., Kolmonovsky, I., and Cesnik, C. E. S., "Controller Agnostic Design Metrics for Stochastic Disturbance Rejection," *AIAA SCITECH Forum*, 2024. doi:10.2514/6.2024-0901, PDF.

[2] Bahia Monteiro, B., Cesnik, C. E. S., and Kolmanovsky, I., "Gust Load Metrics for Multidisciplinary Design Optimization Considering Fatigue and Active Control," *RAeS 8th Aircraft Structural Design Conference*, 2023. PDF.

[3] Bahia Monteiro, B., Gray, A. C., Cesnik, C. E. S., Kolmanovsky, I., and Vetrano, F., "Bi-level Multidisciplinary Design Optimization of a Wing Considering Maneuver Load Alleviation and Flutter," *AIAA SCITECH 2023 Forum*, 2023, p. 0728. doi:10.2514/6.2023-0728, PDF.

[4] Jonsson, E., Riso, C., Monteiro, B. B., Gray, A. C., Martins, J. R. R. A., and Cesnik, C. E. S., "High-Fidelity Gradient-Based Wing Structural Optimization Including Geometrically Nonlinear Flutter Constraint," *AIAA Journal*, Vol. 61, No. 7, 2023, pp. 3045–3061. doi:10.2514/1.J061575, PDF.

HONOURS AND AWARDS

Gold medal award for best GPA of the class of 2019 at UFMG
2nd overall place on ENEM 2013 (Brazilian version of the SAT) — more than 7 million students applied
Gold medal at the XV Brazilian Olympiad of Astronomy and Astronautics (October 2012)
Honorable mention on the state's Olympiad of Mathematics (MG, 2007)

COMPUTER SKILLS

C/C++, Python, Java, Fortran, Git, Matlab/Simulink, NASA's OpenMDAO, Linux/Unix (real-time)

LANGUAGE SKILLS Portuguese (native), English (near-native), Spanish, French, German (beginner)

HOBBIES Glider pilot, mountain biking, tennis, road cycling, skiing, climbing, rowing